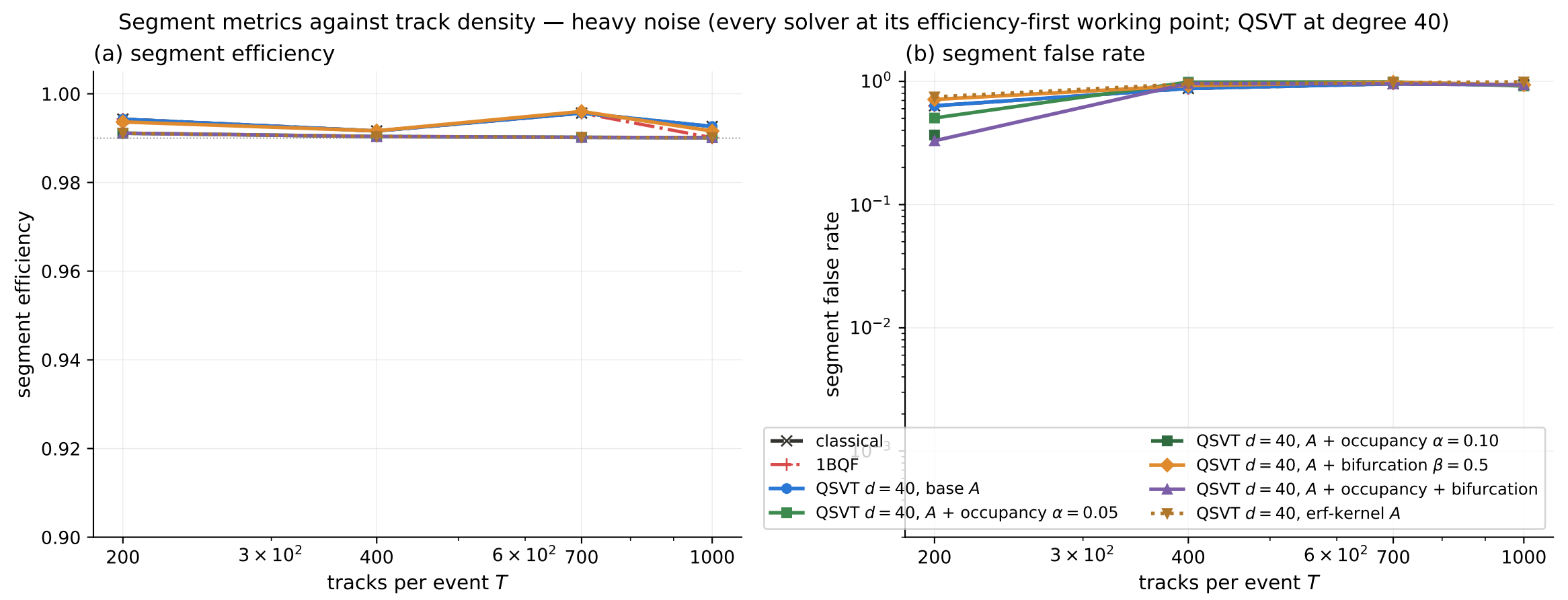

$\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20\,\mu$m, drop 1% · formula $\varepsilon = 6.31$ mrad · $\gamma = 3$, $\delta = 1$, step kernel · occupancy $\alpha = 0.05$ · bifurcation $\beta = 0.5$, $\varepsilon_B = \varepsilon$ · QSVT polynomial: degree 40, coefficients least-squares fit per Hamiltonian in Chebyshev-moment space (ridge weight chosen for best efficiency at false rate $\leq$ 1%) · every solver at its efficiency-first working point: the lowest threshold keeping 99% of the true segments · T reps: 10 (200) / 5 (400) / 3 (700, 1000) · 1BQF = matched cosine $|\cos(\pi\lambda/2s')|$ on the same events